

Euclid, Book I, Proposition I.10

To Bisect a Given Finite Straight Line

CGJteamLab

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1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line.

In Euclid's development this construction depends on several earlier results. An equilateral triangle is first erected on the given segment using Proposition I.1. The vertex angle of this triangle is then bisected by Proposition I.9, and the Side–Angle–Side congruence criterion of Proposition I.4 is used to prove that the intersection point is the midpoint of the original segment.

The Hilbert reconstruction follows a fundamentally different approach. The existence of a midpoint has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this previously derived result.

This proposition therefore provides one of the clearest examples of the difference between Euclid's historical development and a modern axiomatic reconstruction.

2 The Classical Configuration

Let AB be the given finite straight line.

The objective is to construct a point lying on the segment AB that divides it into two congruent parts.

Such a point is called the midpoint of the segment.



Figure 1: The given finite straight line AB .

2.1 Construction

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Next, bisect the angle $\angle ACB$ by Proposition I.9.

Let the angle bisector meet the segment AB at the point D .

The claim is that D is the midpoint of the segment AB .

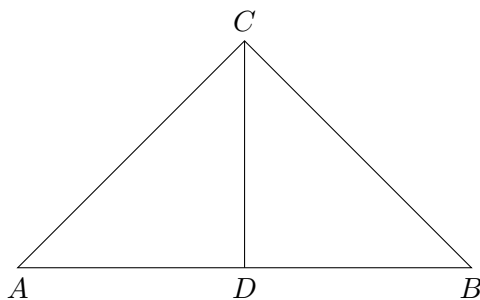


Figure 2: Euclid's construction of the midpoint of a segment.

2.2 Application of Proposition I.4

Consider the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD.$$

Since the triangle ABC is equilateral,

$$AC \cong BC.$$

The segment

$$CD$$

is common to both triangles, and by construction,

$$\angle ACD \cong \angle DCB.$$

Therefore the hypotheses of Proposition I.4 are satisfied.

It follows that

$$AD \cong DB.$$

Since the point D lies on the segment AB , it is the midpoint of the given finite straight line.

3 Statement

Theorem 1 (Euclid I.10). *Let AB be a finite straight line with*

$$A \neq B.$$

Then there exists a midpoint of the segment AB . That is, there exists a point M such that

$$A, M, B$$

are collinear,

$$M$$

lies between A and B , and

$$AM \cong MB.$$

Equivalently, every nondegenerate segment admits a midpoint.

4 Classical Proof

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Bisect the angle $\angle ACB$ by Proposition I.9, and let the bisecting ray meet the segment AB at the point D .

Since

$$AC \cong BC,$$

$$CD \cong CD,$$

and

$$\angle ACD \cong \angle DCB,$$

the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD$$

satisfy the hypotheses of Proposition I.4.

Hence

$$AD \cong DB.$$

Because the point D lies on the segment AB , it is the midpoint of the given finite straight line.

This completes the construction.

5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces the entire classical construction by a previously established theorem of the Hilbert interface.

Rather than constructing an auxiliary equilateral triangle, bisecting an angle, and applying the Side–Angle–Side criterion, the existence of a midpoint is obtained directly from the general midpoint existence theorem.

Consequently, Proposition I.10 becomes a simple wrapper around an existing interface theorem.

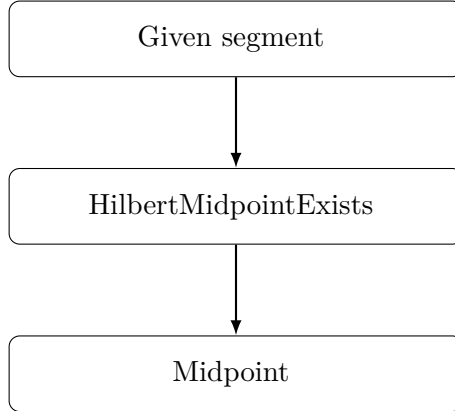


Figure 3: Architecture of the Hilbert reconstruction of Proposition I.10.

6 Hilbert Reconstruction

The reconstruction consists of a direct application of the general midpoint existence theorem.

Given a nondegenerate segment, the theorem `HilbertMidpointExists` immediately produces a point lying between the endpoints and equidistant from them.

No auxiliary geometric construction is required, since the existence of midpoints has already been established independently within the Hilbert development.

This represents one of the largest simplifications obtained by the layered architecture of the Hilbert interface.

7 Lean Formalization

The Lean implementation mirrors the Hilbert reconstruction exactly.

After assuming that the endpoints of the given segment are distinct, the proof consists of a single application of the theorem

`HilbertMidpointExists`.

This theorem immediately returns a point lying between the endpoints and equidistant from them, precisely satisfying the definition of a midpoint.

Consequently, the formal proof contains no auxiliary constructions and makes no explicit use of Propositions I.1, I.4, or I.9. All of the classical geometric reasoning has already been incorporated into the previously established Hilbert interface.

8 Relationship with Earlier Propositions

8.1 Proposition I.1

Euclid's construction of the midpoint begins by constructing an equilateral triangle on the given segment. Proposition I.1 therefore supplies the first auxiliary point and the equal sides needed later in the congruence argument.

8.2 Proposition I.4

The decisive classical step is SAS. After the angle at the apex has been bisected, the two resulting triangles share one side, have a pair of equal sides from the equilateral construction, and have congruent included angles. Proposition I.4 then yields congruence of the two triangles and hence equality of the two halves of the original segment.

8.3 Proposition I.9

Proposition I.9 is the immediate constructional input: it bisects the angle at the apex of the equilateral triangle.

Thus Euclid's dependency chain is particularly transparent:

$$\text{I.1} \longrightarrow \text{I.9} \longrightarrow \text{I.4} \longrightarrow \text{I.10}.$$

8.4 Transition to Proposition I.11

The midpoint construction becomes useful immediately in the classical proof of Proposition I.11, where a symmetric configuration about a point on a line is used to construct a perpendicular.

More generally, midpoint and segment-bisection infrastructure becomes a recurring component of later Book I constructions.

9 Hilbert and Book Zero Components Used

9.1 Midpoint Representation

The formal conclusion is expressed by the project midpoint predicate: the constructed point lies between the endpoints and the two resulting segments are congruent.

Thus midpoint is not treated as an affine or coordinate notion. It is entirely synthetic:

$$A - M - B \quad \text{and} \quad AM \cong MB.$$

9.2 Angle-Bisector Infrastructure

The reconstruction uses the already established angle-bisector construction corresponding to Proposition I.9. This provides the ray needed to divide the auxiliary triangle into two congruent

triangles.

9.3 SAS

The triangle comparison is carried by the Hilbert SAS infrastructure. Once the two triangles are congruent, the required segment congruence follows as a corresponding-side consequence.

9.4 Betweenness and Incidence

The final midpoint statement requires more than segment congruence: the point must lie between the original endpoints. The order layer supplies this positional information explicitly rather than reading it from the diagram.

10 Formalization Notes

Proposition I.10 provides one of the clearest illustrations of the difference between Euclid's historical development and a layered axiomatic reconstruction.

In the *Elements*, the midpoint construction depends on three earlier propositions. Euclid first constructs an equilateral triangle, then bisects one of its angles, and finally applies the Side–Angle–Side criterion to prove that the resulting point divides the original segment into two equal parts.

The Hilbert reconstruction is fundamentally different. The existence of midpoints has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this theorem, requiring no additional geometric construction.

This proposition demonstrates the principal advantage of the layered architecture developed in CGJteamLab. Once sufficiently rich interface theorems have been established, many classical propositions cease to be independent proofs and instead become concise applications of the existing theory.

Proposition I.10 is perhaps the simplest example of this philosophy, reducing an entire classical construction to a single theorem call.

11 Dependency Summary

The classical dependency structure is

$$\begin{array}{c}
\text{given segment } AB \\
\downarrow \\
\text{I.1 : equilateral triangle} \\
\downarrow \\
\text{I.9 : bisect the apex angle} \\
\downarrow \\
\text{I.4 : SAS} \\
\downarrow \\
AM \cong MB \\
\downarrow \\
A - M - B \\
\downarrow \\
\text{euclid_proposition_10.}
\end{array}$$

The Hilbert reconstruction retains the same synthetic ingredients but makes the order component of the midpoint conclusion explicit.

No numerical length, coordinate system, or new geometric axiom is introduced.

12 Conclusion

Proposition I.10 completes the first ten propositions of Book I with the construction of a midpoint.

The classical proof is a compact synthesis of three earlier results: I.1 supplies an equilateral triangle, I.9 bisects its apex angle, and I.4 converts the resulting SAS configuration into equality of the two halves of the given segment.

The formal reconstruction preserves this synthetic character while making one feature especially explicit: a midpoint is not merely a point with two equal distances. Its position between the endpoints is part of the theorem and must be carried by the Hilbert order structure.

I.10 therefore consolidates the early construction and congruence machinery into a reusable segment-bisection result. This midpoint infrastructure becomes immediately relevant to the perpendicular constructions beginning with Proposition I.11.